MAT 101: Linear Algebra I

Module 1: Matrices

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1 What is a Matrix?

1.1 Definition

Definition: Matrix

A **matrix** is a rectangular table of numbers arranged in **rows** (going across) and **columns** (going down), written inside square brackets. The numbers inside are called **entries** (or **elements**).

Example:

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 0 & 4 \end{bmatrix}$$

- Row 1 is $2, 3, 5$. Row 2 is $1, 0, 4$.
- Column 1 is $2, 1$. Column 2 is $3, 0$. Column 3 is $5, 4$.

Matrices are usually named with capital letters: $A, B, C, \ldots$

1.2 Order (size) of a matrix

Definition: Order

If a matrix has m rows and n columns, its **order** is $m \times n$ (read "m by n"). **Always say rows first, then columns.**

Example 1: A above has 2 rows and 3 columns, so its order is 2×3.

Example 2: $B = \begin{bmatrix} 4 \\ 7 \\ 9 \end{bmatrix}$ has 3 rows and 1 column, so its order is 3×1.

Example 3: $C = \begin{bmatrix} 6 & 1 \end{bmatrix}$ has 1 row and 2 columns, so its order is 1×2.

The **total number of entries** is $m \times n$. A 2×3 matrix has $2 \times 3 = 6$ entries.

1.3 Naming the entries

Definition: Entry a_{ij}

The entry in **row** i and **column** j of matrix A is written a_{ij}. The first small number is the **row**. The second is the **column**.

A general $m \times n$ matrix looks like this:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Example: For $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 0 & 4 \end{bmatrix}$:

- a_{12} = row 1, column 2 = 3
- a_{21} = row 2, column 1 = 1
- a_{23} = row 2, column 3 = 4

1.4 Building a matrix from a rule

Sometimes we are given a rule for a_{ij} and asked to write the matrix.

Example: Write the 2×3 matrix where $a_{ij} = i + j$.

Step 1: A 2×3 matrix has rows $i = 1, 2$ and columns $j = 1, 2, 3$.

Step 2: Work out each entry:

	$j = 1$	$j = 2$	$j = 3$
$i = 1$	$1 + 1 = 2$	$1 + 2 = 3$	$1 + 3 = 4$
$i = 2$	$2 + 1 = 3$	$2 + 2 = 4$	$2 + 3 = 5$

Step 3: Write the matrix:

$$\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

1.5 Equal matrices

> **Definition: Equality of matrices**
>
> Two matrices are **equal** if (1) they have the **same order**, and (2) **every pair of matching entries is equal**.

Example: If $\begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$, then matching entries give $x = 5$ and $y = -1$.

1.5.1 Practice Questions 1

1. State the order of each matrix:
   a. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$
   b. $\begin{bmatrix} 7 & 8 & 9 & 0 \end{bmatrix}$
   c. $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$
2. How many entries does a 4×5 matrix have?
3. For $A = \begin{bmatrix} 3 & -1 & 0 & 5 \\ 2 & 4 & 7 & 1 \end{bmatrix}$, find a_{13}, a_{21} and a_{24}.
4. Write the 2×3 matrix with $a_{ij} = i \times j$.
5. Write the 3×3 matrix with $a_{ij} = i - j$.
6. Find a and b if $\begin{bmatrix} a+b & 2 \\ 3 & a-b \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}$.

2 Special Types of Matrices

Some matrices have a special shape or pattern, so they have special names. Learn these names well. They appear in every module of this course.

2.1 Row matrix and column matrix

Definitions

- A **row matrix** has only **one row** (order $1 \times n$).
- A **column matrix** has only **one column** (order $m \times 1$).

Examples: $\begin{bmatrix} 2 & 5 & -1 \end{bmatrix}$ is a row matrix. $\begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$ is a column matrix.

2.2 Zero matrix

Definition: Zero matrix

A **zero matrix** (written O or 0) has **every entry equal to 0**. It can have any order.

Example: The 2×3 zero matrix is $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

2.3 Square matrix and the main diagonal

Definitions

- A **square matrix** has the **same number of rows and columns** ($n \times n$).
- The **main diagonal** of a square matrix is the line of entries $a_{11}, a_{22}, a_{33}, \ldots$ going from top-left to bottom-right.

Example: In $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, the main diagonal is $1, 5, 9$.

Most of the special types below are square matrices.

2.4 Diagonal matrix

Definition: Diagonal matrix

A **diagonal matrix** is a square matrix in which **every entry off the main diagonal is 0**. (The diagonal entries can be anything, even 0.)

Examples:

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix} \qquad \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$$

2.5 Scalar matrix and identity matrix

Definitions

- A **scalar matrix** is a diagonal matrix whose diagonal entries are **all the same number**.
- The **identity matrix** I (or I_n) is the scalar matrix whose diagonal entries are all **1**.

Examples:

$$\text{Scalar: } \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \qquad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The identity matrix acts like the number 1: for any matrix A of a suitable size, $AI = IA = A$. (You will see this in Idea 3.)

2.6 Triangular matrices

Definitions

- **Upper triangular:** a square matrix with **all entries below the main diagonal equal to 0**.
- **Lower triangular:** a square matrix with **all entries above the main diagonal equal to 0**.

Examples:

$$\text{Upper: } \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \qquad \text{Lower: } \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix}$$

Every diagonal matrix is both upper and lower triangular.

2.7 Transpose (needed for the next three types)

> **Definition: Transpose**
>
> The **transpose** of A, written A^T, is found by **turning each row into a column**. Row 1 becomes column 1, row 2 becomes column 2, and so on. If A is $m \times n$, then A^T is $n \times m$. Also $(A^T)_{ij} = a_{ji}$.

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \implies A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

(A 2×3 matrix has a 3×2 transpose.)

2.8 Symmetric matrix

> **Definition: Symmetric matrix**
>
> A square matrix A is **symmetric** if $A^T = A$. This means $a_{ij} = a_{ji}$: the matrix is a **mirror image across the main diagonal**.

Example:

$$\begin{bmatrix} 1 & 5 & 2 \\ 5 & 3 & 7 \\ 2 & 7 & 9 \end{bmatrix}$$

Check: $a_{12} = 5 = a_{21}$, $a_{13} = 2 = a_{31}$, $a_{23} = 7 = a_{32}$.

2.9 Skew-symmetric matrix

Definition: Skew-symmetric matrix

A square matrix A is **skew-symmetric** if $A^T = -A$. This means $a_{ij} = -a_{ji}$. **All diagonal entries must be 0**, because $a_{ii} = -a_{ii}$ is only true when $a_{ii} = 0$.

Example:

$$\begin{bmatrix} 0 & 4 & -1 \\ -4 & 0 & 3 \\ 1 & -3 & 0 \end{bmatrix}$$

Check: $a_{12} = 4$ and $a_{21} = -4$. $a_{13} = -1$ and $a_{31} = 1$. $a_{23} = 3$ and $a_{32} = -3$. Every pair is opposite.

2.10 Orthogonal matrix

Definition: Orthogonal matrix

A square matrix A is **orthogonal** if

$$A^T A = AA^T = I$$

That is, its transpose is also its inverse: $A^T = A^{-1}$.

Example 1: Let $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and

$$A^T A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

So A is orthogonal.

Example 2: $B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ gives $B^T B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \neq I$. So B is **not** orthogonal.

2.10.1 Practice Questions 2

1. Write down (a) the 2×3 zero matrix and (b) the 3×3 identity matrix.
2. Classify each matrix. Choose from: zero, identity, diagonal, upper triangular, lower triangular, symmetric, skew-symmetric. (More than one may apply.)

   a. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

   b. $\begin{bmatrix} 2 & 5 \\ 0 & 3 \end{bmatrix}$

   c. $\begin{bmatrix} 1 & 0 \\ 7 & 2 \end{bmatrix}$

   d. $\begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$

   e. $\begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

3. Find the transpose of $\begin{bmatrix} 1 & -2 \\ 3 & 0 \\ 5 & 4 \end{bmatrix}$.
4. Find x so that $\begin{bmatrix} 1 & x \\ 4 & 2 \end{bmatrix}$ is symmetric.
5. Find a, b, c so that $\begin{bmatrix} 0 & a & 2 \\ 3 & 0 & b \\ c & -5 & 0 \end{bmatrix}$ is skew-symmetric.
6. Is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ orthogonal? Show your working.
7. Is $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ orthogonal? Why or why not?

3 Matrix Operations

3.1 Addition and subtraction

> **i** Rule: Adding and subtracting matrices
>
> You can add (or subtract) two matrices **only if they have the same order**. To do it, add (or subtract) the **matching entries**.

Example 1 (addition):

$$\begin{bmatrix} 2 & 5 & 1 \\ 0 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & -2 & 6 \\ 7 & 3 & -1 \end{bmatrix} = \begin{bmatrix} 2+1 & 5+(-2) & 1+6 \\ 0+7 & -3+3 & 4+(-1) \end{bmatrix} = \begin{bmatrix} 3 & 3 & 7 \\ 7 & 0 & 3 \end{bmatrix}$$

Example 2 (subtraction): Using the same two matrices,

$$\begin{bmatrix} 2 & 5 & 1 \\ 0 & -3 & 4 \end{bmatrix} - \begin{bmatrix} 1 & -2 & 6 \\ 7 & 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 7 & -5 \\ -7 & -6 & 5 \end{bmatrix}$$

Example 3 (not possible): A 2×3 matrix and a 3×2 matrix have different orders, so they **cannot** be added.

Rules that work (for matrices of the same order):

- $A + B = B + A$ (order does not matter)
- $(A + B) + C = A + (B + C)$
- $A + O = A$ (adding the zero matrix changes nothing)

3.1.1 Practice Questions 3A

Let $P = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $Q = \begin{bmatrix} 0 & -1 \\ 5 & 2 \end{bmatrix}$.

1. Find $P + Q$.
2. Find $P - Q$ and $Q - P$. Are they the same?
3. Can you add $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$? Explain.
4. Find the matrix X such that $P + X = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$.

3.2 Scalar multiplication

> **i** Rule: Scalar multiplication
>
> A **scalar** is an ordinary number. To multiply a matrix by a scalar k, **multiply every entry by** k.

Example 1:

$$4 \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 4 & -8 \\ 0 & 12 \end{bmatrix}$$

Example 2: (fractions are allowed)

$$\tfrac{1}{2}\begin{bmatrix} 4 & 6 \\ -2 & 10 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$$

Example 3 (combining operations): Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$. Find $3A - 2B$.

Step 1: $3A = \begin{bmatrix} 3 & 6 \\ 0 & 3 \end{bmatrix}$

Step 2: $2B = \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix}$

Step 3: Subtract: $3A - 2B = \begin{bmatrix} 3-4 & 6-0 \\ 0-2 & 3-2 \end{bmatrix} = \begin{bmatrix} -1 & 6 \\ -2 & 1 \end{bmatrix}$

3.2.1 Practice Questions 3B

1. Find $-2\begin{bmatrix} 1 & -3 \\ 4 & 0 \end{bmatrix}$.
2. Find $\frac{1}{3}\begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$.
3. With P and Q from Practice 3A, find $2P + 3Q$.
4. With P and Q from Practice 3A, find $P - 2Q$.

3.3 Matrix multiplication

This is the most important operation, so we go slowly.

3.3.1 Step 1: When is multiplication possible?

> **i** Rule: Can A and B be multiplied?
>
> AB exists only if the **number of columns of** A equals the **number of rows of** B. If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

A quick way to check, write the orders side by side:

$(m\times\underline{n})\ (\underline{n}\times p) \longrightarrow$ the two middle numbers must match; the outer numbers give the answer's order: $m\times p$

Examples:

- $(2 \times 3)(3 \times 4)$: middle numbers match (3 and 3). Answer is 2×4.
- $(3 \times 2)(3 \times 2)$: middle numbers are 2 and 3. **Not possible.**
- $(1 \times 4)(4 \times 1)$: answer is 1×1.
- $(4 \times 1)(1 \times 4)$: answer is 4×4.

3.3.2 Step 2: Row times column

To find the entry in **row** i, column j of AB: take **row** i of A and **column** j of B. Multiply the matching entries, then **add** them.

Smallest example (row matrix times column matrix):

$$\begin{bmatrix}1 & 2 & 3\end{bmatrix}\begin{bmatrix}4\\5\\6\end{bmatrix} = \begin{bmatrix}(1)(4)+(2)(5)+(3)(6)\end{bmatrix} = \begin{bmatrix}4+10+18\end{bmatrix} = \begin{bmatrix}32\end{bmatrix}$$

3.3.3 Step 3: A full 2×2 example

Let $A = \begin{bmatrix}2 & 1\\0 & 3\end{bmatrix}$ and $B = \begin{bmatrix}1 & 4\\2 & 5\end{bmatrix}$. Find AB.

- Entry (1,1) = row 1 of A times column 1 of B: $(2)(1)+(1)(2) = 4$
- Entry (1,2) = row 1 of A times column 2 of B: $(2)(4)+(1)(5) = 13$
- Entry (2,1) = row 2 of A times column 1 of B: $(0)(1)+(3)(2) = 6$
- Entry (2,2) = row 2 of A times column 2 of B: $(0)(4)+(3)(5) = 15$

$$AB = \begin{bmatrix}4 & 13\\6 & 15\end{bmatrix}$$

3.3.4 Step 4: A 2×3 times 3×2 example

Let $A = \begin{bmatrix}1 & 0 & 2\\3 & 1 & -1\end{bmatrix}$ and $B = \begin{bmatrix}2 & 1\\0 & 3\\1 & -2\end{bmatrix}$.

Orders: $(2 \times 3)(3 \times 2)$ gives a 2×2 answer.

- (1,1): $(1)(2)+(0)(0)+(2)(1) = 4$
- (1,2): $(1)(1)+(0)(3)+(2)(-2) = -3$
- (2,1): $(3)(2)+(1)(0)+(-1)(1) = 5$

- (2,2): $(3)(1) + (1)(3) + (-1)(-2) = 8$

$$AB = \begin{bmatrix} 4 & -3 \\ 5 & 8 \end{bmatrix}$$

3.3.5 Step 5: Order matters!

Using A and B from Step 3, compute BA:

$$BA = \begin{bmatrix} 1 & 4 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} (1)(2) + (4)(0) & (1)(1) + (4)(3) \\ (2)(2) + (5)(0) & (2)(1) + (5)(3) \end{bmatrix} = \begin{bmatrix} 2 & 13 \\ 4 & 17 \end{bmatrix}$$

But $AB = \begin{bmatrix} 4 & 13 \\ 6 & 15 \end{bmatrix}$. So $AB \neq BA$.

> ⚠ Important facts about multiplication
>
> - In general, $AB \neq BA$. Matrix multiplication is **not commutative**.
> - $AB = O$ does **not** mean $A = O$ or $B = O$. For example $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Rules that do work (when the orders allow the products):

- $(AB)C = A(BC)$
- $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
- $AI = IA = A$
- $AO = OA = O$

3.3.6 Practice Questions 3C

1. Say whether each product exists. If it does, give its order.
   a. $(2 \times 3)(3 \times 4)$
   b. $(3 \times 2)(3 \times 2)$
   c. $(1 \times 4)(4 \times 1)$
   d. $(4 \times 1)(1 \times 4)$
2. Find $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.
3. Find $\begin{bmatrix} 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$.

4. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Find AB and BA.
5. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Show that $AI = A$, where I is the 2×2 identity matrix.
6. Find $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 4 & 0 \end{bmatrix}$.

3.4 Powers of a square matrix

> **Definition: Powers**
>
> For a square matrix A: $A^2 = AA$, $A^3 = AAA$, and so on.

Example: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

$$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad A^3 = A^2 A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

Powers of a **diagonal** matrix are easy: just raise each diagonal entry to the power. For $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $D^2 = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$.

3.4.1 Practice Questions 3D

1. Find A^2 for $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$.
2. Find D^3 for $D = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$.
3. Find A^2 for $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. What do you notice?

3.5 Transpose and its rules

We defined the transpose in Idea 2. Here are its rules.

Rules for the transpose

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(kA)^T = kA^T$
4. $(AB)^T = B^T A^T$ (**the order reverses!**)

Example (checking Rule 4): Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

$AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$.

$B^T A^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$. They match.

3.5.1 Practice Questions 3E

1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Find $A^T A$. Is it symmetric?
2. Let $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$. Verify that $(AB)^T = B^T A^T$.
3. If A is 3×2, what is the order of A^T? What is the order of $A^T A$?

4 Elementary Row Operations

4.1 Definition

Definition: Elementary row operations (EROs)

There are exactly **three** elementary row operations:

1. **Swap** two rows: $R_i \leftrightarrow R_j$
2. **Scale** a row by a nonzero number k: $R_i \to kR_i$
3. **Add** a multiple of one row to another row: $R_i \to R_i + kR_j$

Why do we use them? They let us simplify a matrix step by step **without changing the information it carries**. This is the engine behind finding inverses (this module) and solving systems of equations (Module 3).

Two matrices are called **row equivalent** if one can be changed into the other using EROs.

4.2 Examples of each operation

Operation 1 (swap): $R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

Operation 2 (scale): $R_1 \to \frac{1}{2}R_1$

$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{bmatrix} 1 & 2 \\ 6 & 8 \end{bmatrix}$$

Operation 3 (add a multiple): $R_2 \to R_2 - 4R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 4R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \end{bmatrix}$$

How the new row 2 was found: $(4 - 4 \cdot 1,\ 5 - 4 \cdot 2,\ 6 - 4 \cdot 3) = (0, -3, -6)$.

> ⚠ Common mistakes
>
> - In $R_2 \to R_2 - 4R_1$, **only row 2 changes**. Row 1 stays the same.
> - You may **not** scale a row by 0.
> - Do one operation at a time and write down which one you used.

4.3 Every operation can be undone

- Swap $R_i \leftrightarrow R_j$ is undone by swapping again.
- $R_i \to kR_i$ is undone by $R_i \to \frac{1}{k}R_i$.
- $R_i \to R_i + kR_j$ is undone by $R_i \to R_i - kR_j$.

4.3.1 Practice Questions 4

1. For $M = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 1 & 0 & 8 \end{bmatrix}$, apply $R_2 \to R_2 - 2R_1$. Write the new matrix.
2. On your result from Question 1, apply $R_3 \to R_3 - R_1$.
3. Which operation changes $\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ into $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$?

4. Which operation changes $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ into $\begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$?
5. Apply $R_1 \leftrightarrow R_3$ to $\begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 7 & 9 \end{bmatrix}$.
6. What operation undoes $R_3 \to R_3 + 5R_1$?

5 Row Echelon Form

5.1 Definitions

First, two small terms:

- A **zero row** is a row where every entry is 0.
- The **leading entry** (or **pivot**) of a nonzero row is its **first nonzero entry from the left**.

i Definition: Row echelon form (REF)

A matrix is in **row echelon form** if:

1. All **zero rows are at the bottom**.
2. Each pivot is **to the right** of the pivot in the row above it (a "staircase" going down to the right).
3. All entries **below** each pivot are 0.
4. Each pivot is **1**. (Some books do not require this. We use it in this course, because it makes later steps easier.)

i Definition: Reduced row echelon form (RREF)

A matrix is in **reduced row echelon form** if it is in REF **and** each pivot is the **only nonzero entry in its column** (zeros above the pivots too).

5.2 Recognising REF and RREF

Matrix	REF?	RREF?	Reason
$\begin{bmatrix}1 & 2 & 3\\0 & 1 & 4\\0 & 0 & 1\end{bmatrix}$	Yes	No	Staircase is correct, but there are nonzero entries above the pivots
$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$	Yes	Yes	Identity matrix
$\begin{bmatrix}1 & 0 & 2\\0 & 1 & 3\\0 & 0 & 0\end{bmatrix}$	Yes	Yes	Zero row at the bottom; pivot columns have zeros elsewhere
$\begin{bmatrix}1 & 2\\0 & 0\\0 & 3\end{bmatrix}$	No	No	Zero row is not at the bottom
$\begin{bmatrix}0 & 1 & 2\\1 & 0 & 3\end{bmatrix}$	No	No	Pivot of row 2 is to the **left** of the pivot of row 1
$\begin{bmatrix}2 & 1\\0 & 1\end{bmatrix}$	No	No	First pivot is 2, not 1

5.3 How to reduce a matrix to REF (the method)

Work **column by column, from left to right**:

1. Find the first column that has a nonzero entry. Use a swap if needed to bring a nonzero entry to the top of the unfinished rows.
2. Scale that row to make the pivot **1**.
3. Use "add a multiple" operations to make all entries **below** the pivot **0**.
4. Ignore the finished row and column. Repeat on what remains.

To go on to RREF, then work **upwards**: use each pivot to make the entries **above** it 0.

5.3.1 Example A (a 2×2 matrix)

Reduce $\begin{bmatrix}2 & 4\\3 & 5\end{bmatrix}$.

Step 1: Make the first pivot 1: $R_1 \to \frac{1}{2}R_1$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

Step 2: Clear below it: $R_2 \to R_2 - 3R_1$

$$\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

Step 3: Make the second pivot 1: $R_2 \to -R_2$

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \text{(this is REF)}$$

Step 4 (for RREF): Clear above the second pivot: $R_1 \to R_1 - 2R_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{(this is RREF)}$$

5.3.2 Example B (a 3×3 matrix)

Reduce $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ -1 & 0 & 2 \end{bmatrix}$.

Step 1: Clear below the first pivot. $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 + R_1$:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 2 & 1 \end{bmatrix}$$

Step 2: Make the second pivot 1: $R_2 \to -R_2$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 2 & 1 \end{bmatrix}$$

Step 3: Clear below it: $R_3 \to R_3 - 2R_2$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 7 \end{bmatrix}$$

Step 4: Make the third pivot 1: $R_3 \to \frac{1}{7}R_3$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{(REF)}$$

Step 5 (for RREF): Work upwards. $R_2 \to R_2 + 3R_3$ and $R_1 \to R_1 + R_3$:

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then $R_1 \to R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{(RREF)}$$

5.3.3 Example C (a matrix that produces a zero row)

Reduce $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$.

Step 1: $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

Step 2: The zero row must go to the bottom: $R_2 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3: $R_2 \to -R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{(REF)}$$

Step 4 (for RREF): $R_1 \to R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{(RREF)}$$

5.4 Rank (a useful by-product)

> **Definition: Rank**
>
> The **rank** of a matrix is the **number of nonzero rows** in its row echelon form.

In Example B the rank is 3. In Example C the rank is 2 (one zero row).

> **Note**
>
> REF is **not unique**. Different (correct) sequences of operations can give different REFs. The **RREF**, however, is always the same for a given matrix.

5.4.1 Practice Questions 5

1. Which of these are in REF? Which are in RREF?
   a. $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$
   b. $\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \end{bmatrix}$
   c. $\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$
   d. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
2. Reduce $\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ to REF, then to RREF.
3. Reduce $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ to RREF.
4. Reduce $\begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 8 \\ 0 & 0 & 5 \end{bmatrix}$ to REF.

5. Reduce $\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 2 & 3 & 4\end{bmatrix}$ to RREF and state its rank.
6. Reduce $\begin{bmatrix}2 & 4 & 2\\ 1 & 3 & 4\end{bmatrix}$ to REF.

6 The Inverse of a Matrix

6.1 Definition

> **Definition: Inverse matrix**
>
> Let A be a **square** matrix. If there is a matrix A^{-1} of the same order such that
>
> $$AA^{-1} = A^{-1}A = I$$
>
> then A^{-1} is called the **inverse** of A, and A is called **invertible** (or **non-singular**). If no such matrix exists, A is **singular**.

Think of it like numbers: the inverse of 5 is $\frac{1}{5}$, because $5 \times \frac{1}{5} = 1$. For matrices, "1" is the identity matrix I.

Example (checking an inverse): Let $A = \begin{bmatrix}1 & 2\\ 3 & 7\end{bmatrix}$ and $B = \begin{bmatrix}7 & -2\\ -3 & 1\end{bmatrix}$.

$$AB = \begin{bmatrix}(1)(7)+(2)(-3) & (1)(-2)+(2)(1)\\ (3)(7)+(7)(-3) & (3)(-2)+(7)(1)\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = I$$

So $B = A^{-1}$.

Only square matrices can have inverses, and an inverse (if it exists) is **unique**.

6.2 Method: row reduction on $[A \mid I]$

Idea: Write A and I side by side. Use EROs to turn the **left side** into I. The same operations turn the **right side** into A^{-1}.

$$[A \mid I] \xrightarrow{\text{EROs}} [I \mid A^{-1}]$$

Steps:

1. Write the augmented matrix $[A \mid I]$.

2. Work column by column (as in Idea 5) to get 1s on the diagonal of the left side and 0s everywhere else on the left side.
3. Do every operation on the **whole row** (both sides).
4. When the left side is I, the right side is A^{-1}.

6.2.1 Example 1 (2×2)

Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$.

Step 1: Set up:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right]$$

Step 2: $R_2 \to R_2 - 3R_1$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right]$$

Step 3: $R_1 \to R_1 - 2R_2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right]$$

So $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. This matches the check in 6.1.

6.2.2 Example 2 (2×2 needing a scaling step)

Find the inverse of $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_2} \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

Check: $AA^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2-1 & -2+2 \\ 1-1 & -1+2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

6.2.3 Example 3 (3×3)

Find the inverse of $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

Step 1: Set up:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

Step 2: $R_3 \to R_3 - R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array}\right]$$

Step 3: $R_3 \to R_3 - R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right]$$

Step 4: $R_3 \to -\frac{1}{2}R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right]$$

Step 5: Clear above the last pivot: $R_1 \to R_1 - 2R_3$ and $R_2 \to R_2 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

Quick check of one entry: Row 1 of A times column 1 of A^{-1} is $(1)(0)+(0)(-\frac{1}{2})+(2)(\frac{1}{2}) = 1$.

6.3 When there is no inverse

> ⚠ Singular matrices
>
> If, while reducing $[A \mid I]$, the **left side gets a row of all zeros**, then A has **no inverse**. A is singular.

Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right]$$

The left side has a zero row, so no inverse exists. (Row 2 of A is 2 times row 1, so the rows are not independent.)

6.4 Useful rules for inverses

Assume A and B are invertible matrices of the same order.

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$ (**the order reverses**, like the transpose)
3. $(A^T)^{-1} = (A^{-1})^T$
4. The inverse of a diagonal matrix (with no zero on the diagonal) is found by inverting each diagonal entry. For example, $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{5} \end{bmatrix}$.

6.4.1 Practice Questions 6

1. Show that $B = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ is the inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$.
2. Find the inverse of $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ by row reduction. Compare with Question 1.
3. Find the inverse of $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$.
4. Does $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ have an inverse? Show why or why not.

5. Find the inverse of $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ without row reduction.
6. Find the inverse of $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.
7. If $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$, what is A?

7 Tutorial Questions — Module 1

Attempt all questions. Show your working. Answers are at the end.

Section A: Basics and special matrices

1. State the order of $A = \begin{bmatrix} 4 & -1 & 0 \\ 2 & 5 & 7 \end{bmatrix}$ and write down a_{13} and a_{22}.
2. Write the 3×3 matrix with $a_{ij} = 2i - j$.
3. Classify each matrix (diagonal, identity, upper triangular, lower triangular, symmetric, skew-symmetric, orthogonal, or none):

   a. $\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}$

   b. $\begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix}$

   c. $\begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & 5 & 6 \end{bmatrix}$

   d. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

4. Find a so that $\begin{bmatrix} 2 & a \\ 5 & 1 \end{bmatrix}$ is symmetric.
5. Show that $\begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix}$ is orthogonal.

Section B: Operations

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 4 & 0 \end{bmatrix}$.

6. Find $A + B$ and $A - B$.
7. Find $2A - 3B$.
8. Find AB and BA. Are they equal?
9. Find A^2.

10. Verify that $(AB)^T = B^T A^T$ for these matrices.

Section C: Row operations and echelon form

11. Apply $R_2 \to R_2 - 3R_1$ then $R_3 \to R_3 - 2R_1$ to $\begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & 5 & 1 \end{bmatrix}$.
12. Reduce $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 1 & 2 \end{bmatrix}$ to REF and then to RREF. State its rank.
13. Reduce $\begin{bmatrix} 0 & 2 & 4 \\ 1 & 1 & 1 \end{bmatrix}$ to REF.

Section D: Inverses

14. Find the inverse of $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$ by row reduction and check your answer.
15. Find the inverse of $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.
16. Show that $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ has no inverse.

Section E: Proofs

17. Prove that if A is any square matrix, then $A + A^T$ is symmetric.
18. Prove that if A and B are symmetric and $AB = BA$, then AB is symmetric.
19. Prove that if A and B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$.

End of Module 1. Next: Module 2 — Determinants.